DR. OWEN L. MARTIN

Research Scientist, Asst — University of Wyoming — WyldTech

Broadly, I study animal communication and how individual variability influences collective behavior within communicating groups. During my Ph.D, I studied synchronization, behavioral phenology, and visual signal differentiation in the *Lampyridae* family (firefly beetles). Currently, I study wildlife migration across the American West with computational tools. I also develop and deploy software for threatened species population monitoring applications.

EDUCATION

Ph.D in Computer Science, University of Colorado Boulder — August 2020 - December 2025
Research assistant to Dr. Orit Peleg
Dissertation title: On Synchronization and Distinguishability in Firefly Flash Behavior

Tropical Bioacoustics Certificate, University of St. Etienne — May 2025
Area of Concentration: Marine Bioacoustics at CRIOBE station in Mo'orea, French Polynesia

M.S. in Computer Science, University of Colorado Boulder — August 2020 - December 2023
Area of Concentration: Complex Systems

B.S. in Computer Science, Tufts University — August 2013 - May 2017

RESEARCH PROJECTS

Pronghorn Abundance in the American West — Ongoing
University of Wyoming

- Aerial wildlife survey CLI integrating GPS metadata extraction, viewshed computation, and pronghorn detection via PyTorch
- Geospatial pipeline with rasterio, geopandas, pyproj to orthorectify image footprints and analyze coverage redundancy across survey regions
- Automated S3 data ingestion and interactive map generation

Wyoming Soundscape Biodiversity — Ongoing
University of Wyoming

- Camera trap and bioacoustic recorder data analysis assessing biodiversity and behavior of threatened species within sagebrush steppe ecosystems
- Fine-tuned bioacoustic pipeline with librosa, scikit-maad, opensoundscape, AVES, and NatureLM for audio detections
- Fine-tuned camera trap pipeline with MegaDetector + BioCLIP2 for visual detections

I-80 Wildlife Crossing Monitoring — Ongoing
University of Wyoming

- Real-time camera trap and acoustic data processing using the same models on data gathered from underpasses along I80 in Wyoming

Thesis project: Synchronization between LEDs and Fireflies — 2026
University of Colorado Boulder + Congaree National Park, South Carolina

- RaspberryPi + OpenCV integrating observed flash timings with synchronization model
- Designed and conducted experimental assays measuring interactions between fireflies and LED-camera system
- Data analysis with numpy, scipy, pandas to measure firefly phase response

Thesis project: Identifying and Classifying Firefly Species from Flash Patterns — 2024
University of Colorado Boulder

- Organized nationwide collection of flash pattern data and analyzed with stereoscopic computer vision techniques to create dataset of 3D reconstructions of flash pattern behavior
- Developed and trained RNN for species classification by sequence with PyTorch
- Developed unsupervised learning pipeline for flash motif separation and analysis
- Deployed model to Google Cloud Platform in containerized data processing pipeline

Thesis project: Communication Network Reconstruction in Synchronous Firefly Swarms 2026
University of Colorado Boulder

- Designed and conducted experimental assays measuring individual variability within controlled firefly network structure
- Identified nodes with DBSCAN and edges with temporal Bayesian normalized count to produce an inferred network structure
- Analyzed network degree distributions and clustering with networkx

Thesis project: Agent based modeling of firefly synchronization 2023
University of Colorado Boulder + Great Smoky Mountains National Park, Tennessee

- Designed and conducted experimental assays measuring flash pattern variance as it changes with density in controlled firefly swarms
- Programmed and simulated agent-based models of coupled oscillator synchronization to fit to experimental results

PUBLICATIONS

Excitation–inhibition interactions mediate firefly flash synchronization January 2026
***O. Martin**, N. Nechyporenko, K. Jayaram, and O. Peleg*
accepted and in review, iScience. DOI

Behavioral phenology of Colorado Front Range firefly populations revealed by 3D reconstructions of flash patterns October 2024
***O. Martin**, B. Russell, H. Burroughs, R. O'Brien, D. Frances, A. Walker, C. Fallon, and O. Peleg*
In Preparation.

Embracing firefly flash pattern variability with data-driven species classification February 2024
***O. Martin**, C. Nguyen, R. Sarfati, M. Chowdhury, M. Iuzzolino, D.M.T. Nguyen, R. Layer, O. Peleg*
Scientific Reports. DOI

Crowdsourced dataset of firefly trajectories obtained by automated stereo calibration of 360-degree cameras June 2023
*R. Sarfati, **O. Martin**, J.C. Hayes, A.C.S. Owens, P. Shaw, C. Mollohan, R.L. Day, P. C. Mauney, R.V. Joyce, J.Davis, P. Butler, R. Schreiber, B. Auman, and O. Peleg.*
DataDryad. DOI

Emergent periodicity in the collective synchronous flashes of fireflies March 2023
*R. Sarfati, K. Joshi, **O. Martin**, S. Iyer-Biswas, O. Peleg.*
eLife. DOI

TEACHING EXPERIENCE

Graduate Teaching Assistant
University of Colorado Boulder
Data Structures (x3) *Fall 2020, Spring 2021, Spring 2022*

- Taught recitations, introducing linked list, graph, tree, stack, heap, hash table, and queue data structures and associated algorithms

- Wrote and distributed assignment grading software for the TA staff
- Conducted in-person homework grading interviews and office hours
- Designed, produced, and graded quiz material

Senior Capstone Project *Fall 2022, Spring 2025*

- Managed eight engineering teams comprised of senior undergraduate students
- Facilitated communication between teams and sponsor corporations
- Provided technical advice and assistance in application development

Dynamic Models in Biology *Spring 2023*

- Wrote and graded assignments focusing on simulating flocking behavior, random walks, and evolution
- Lectured on biological and computational neurons, neuronal synchronization, and modeling firefly behavior
- Ran homework groups in office hours

Jungle to the Sea in Ecuador *Winter 2023-2024*

- Provided biodiversity expertise on 200+ Amazonian and Galapagan bird species
- Facilitated successful travel for fourteen undergraduates abroad
- Taught hands on field education in evolutionary biology

Undergraduate Teaching Assistant January 2015 - May 2017
Tufts University
Courses taught: Data Structures (x4)

- Wrote and shared assignment grading software for the TA staff
- Conducted in-person homework grading interviews and office hours
- Designed, produced, and graded quiz and exam material

AWARDS

WyldTech Seed Grant April 2026
Assessing ecosystem biodiversity from soundscapes. Fine-tuning Wyoming wildlife detector models from acoustic and camera trap recordings.

NAIRR Startup Research Allocation March 2026
Model training and validation for ungulate detector.

Carol B. Lynch Dissertation Completion Fellowship Fall 2025
University-wide dissertation completion fellowship supporting one semester of graduate research assistant stipend.

Google Cloud Research Credits June 2025
Support for the FlashFinder cloud infrastructure project.

Dynamics Days Travel Award January 2025
Revisiting Winfree's Firefly Machine: Experiments with Synchronous Arrays of Photuris frontalis Fireflies.

Boulder OSMP Research Grant March 2024 - December 2024
From spatiotemporal characterization to ecological insight:
leveraging flash patterns to explore behavioral trends in Colorado Photuris fireflies.
Featured in this article

1st Place, Best Student Talk October 2023
Embracing behavioral variability for data-driven classification of firefly flash patterns
GeoBON Monitoring Biodiversity for Action, Montreal, Canada

Google Cloud Research Credits June 2023
Support for the FlashFinder cloud infrastructure project.

Outstanding Teaching Assistant May 2022
University of Colorado Boulder CS Department

Microsoft PhD Fellowship Nominee May 2022
University of Colorado Boulder CS Department

Apple AIML PhD Fellowship Nominee October 2021
University of Colorado Boulder CS Department

Best DataBlitz Presentation August 2021
ESI Systems Neuroscience Conference

Computer Science Department Professional Development Award May 2021
University of Colorado Boulder

PRESENTATIONS

AI-rial Surveys: Improving Pronghorn Abundance Estimation with
Automated Detection and Self-Correcting Viewshed Projection April 2026
Poster, University of Wyoming Research and Development Symposium
Laramie, WY, USA.

Type-II phase response dynamics facilitate synchronization in *Photuris frontalis* fireflies March 2026
Session talk, American Physical Society Global Summit
Denver, CO, USA.

Unsupervised motif detection in lightscapes September 2025
Invited talk, University of Wyoming Computing Symposium
Laramie, WY, USA.

Entrainment experiments in the synchronous firefly *Photuris frontalis* April 2025
Invited talk, CU Applied Math Seminar Series
Boulder, CO, USA.

FireflAI: Community-Driven Computation Tools for Rare Firefly Population Monitoring March 2025
Invited talk, AI for Animals 2025
Berkeley, CA, USA.

Discovering Flash Pattern Motifs in Fireflies through Unsupervised Learning March 2025
Session talk, American Physical Society Global Summit
Anaheim, CA, USA.

Revisiting Winfree's Firefly Machine:
Experiments with Synchronous Arrays of Photuris frontalis Fireflies January 2025
Poster, Dynamics Days 2025
Denver, CO, USA.

Embracing behavioral variability with data driven classification of firefly flash patterns March 2024
Invited talk, Digital Entomology Symposium, ESA North Central Branch Meeting.
Fort Collins, CO, USA.

Embracing behavioral variability for data-driven classification of firefly flash patterns October 2023
Invited talk, AI for Insect Monitoring Focus Group, GEOBON Monitoring Biodiversity for Action.
Montreal, Canada.

Exploring open- and closed-loop communication between fireflies and LEDs March 2023
Session talk, American Physical Society March Meeting.
Las Vegas, NV, USA.

A pipeline for automatic classification of firefly species from flash patterns Sept 2022
Invited talk, University of Colorado Biofrontiers QED Supergroup.
Boulder, CO, USA.

Exploring Synchronization in Firefly-LED systems April 2022
Session talk, Aspen Center for Physics Dynamics of Living Systems Conference.
Aspen, CO, USA.

Synchronization Dynamics of Firefly-LED systems. March 2022
Session talk, American Physical Society March Meeting,
Minneapolis, MN, USA.

Visual communication of synchronous firefly swarms in natural and virtual realities August 2021
DataBlitz talk, ESI Systems Neuroscience Conference.
Frankfurt, Germany.

LEADERSHIP AND OUTREACH

Graduate Student Mentorship January 2023 - present
Rajath Vasisth, graduate student working on cloud pipeline for firefly data processing
Ayushi Sabnis, graduate student working on improvements to the neural network classification
Jonathan Gorman, graduate student working on cloud pipeline for firefly data processing
Hana Burroughs, undergraduate summer student working on firefly abundance in Colorado
Kumpeerakij Chanin, graduate student rotation working on firefly synchronization modeling
Bowman Russell, undergraduate summer student working on Colorado Front Range firefly populations
Anurag Ranjan, undergraduate student working on firefly synchronization dynamics

Colorado Firefly Project Science Liaison May 2023 - present
City of Boulder Open Space and Mountain Parks,
City of Fort Collins Natural Areas,
City of Loveland Natural Areas
Featured in this article and this article

Computer Science Graduate Student Association Graduate Committee Liaison August 2021 - May 2022
University of Colorado Boulder

Outside Science, Inside Parks April 2022
National Park Service
Featured in this article

Congaree National Park Firefly Education June 2021
National Park Service

SERVICE

Reviewer: MDPI Mathematics April 2026

Organizer: WyldTech + Computing Meetup March 2026

Reviewer: Royal Society Interface July 2022

TECHNICAL SKILLS

Programming Languages	Python, R, Matlab, C++, Javascript, C, Processing, bash, NetLogo
Frameworks	PyTorch, pandas, numpy, rasterio, scipy, tidyverse, dplyr, React, Node, scikit-learn
Tools	Claude Code, Git, Docker, Azure, AzureML, Kubernetes, GCP, slurm, AWS

INDUSTRY EXPERIENCE

Software Engineer in Test NetApp Solidfire December 2018 - May 2020

- Designed and implemented test automation framework for the performance and scaling of enterprise cloud storage solutions, increasing test coverage of virtual networks and unattended deployment features by ninety percent.
- Wrote and deployed dev-ops tooling to facilitate company-wide continuous integration and continuous deployment of features, migrating all tests to containerized execution environment.

- Python, Docker, Jenkins

Software Quality Engineer Hitachi Vantara August 2017 - September 2018

- Designed and tested edge integration features with enterprise cloud storage
- Development of test tracking website